

Assignment 2- Case Study

Stage 3 Lab Activities

SmartCare Domain Modelling

AI OFF -> AI ON -> COMPARE -> VERIFY | 1 hour

A - Requirements Review

Highlight nouns, verbs and business rules in SmartCare v0.2.

B - Candidate Classes – Completed in the tutorial activity

Record candidate concepts, supporting requirements, state and behaviour.

C - CRC Cards – Completed in both the workbook and tutorial activity

Create CRC cards for Patient, Practitioner and Appointment.

D - UML Model – Completed in the workbook

Draw classes, attributes, operations, associations and multiplicities.

E - AI Design Review – Completed in the workbook

Ask AI to suggest classes and relationships using only confirmed requirements; require supporting requirement IDs.

F - Compare and Decide Completed in the tutorial with all being rejected and only one being changed

Record at least one accepted, modified and rejected AI suggestion.

AppointmentManager

G - Python Skeletons. Shown in smartcare_v03.py

Create simple Patient, Practitioner and Appointment class skeletons.

H - Consistency Check

Check model-code consistency; do not implement full behaviour yet.

Reflection

What modelling decision was hardest? Where did AI over-design? What evidence supported your final choices?

Trying to decide what stays apart of a class or attribute. Status and cancellation are a part of appointment rather than a separate class. Also hard to decide if Clinic becomes a class but the requirements didn't confirm it to make it one. AI over designed all the models class suggestions. They are for a larger system, but the requirement was for a small and simple application. I rejected most of the suggestions with only modifying AppointmentManager as it goes under the Appointment class. My final choices were supported by the confirmed requirements, main classes and appointment behaviour. I used the requirement IDs created in week 5 to link it to what the client has asked for with confirmation.